Don't worry if you have a newer version than shown here. You're all set.

Before We Begin

Got Docker installed? Great, you're just in time—I didn't want to start without you. Before we get our hands dirty and start playing with Docker, it's helpful to understand two fundamental concepts: *containers* and *images*.

What Is a Container?

Conceptually, a container is an isolated or "sandboxed" execution environment—an empty vessel for executing software in. Containers rely on virtualization features built in to the Linux (and more recently, Windows[12]) kernel, which let you create a fully isolated set of processes that don't know (or care) about the rest of the system. In fact, inside a container, it appears to be a complete Linux (or Windows) system, even though, in reality, all its resources and capabilities come from the host machine it's running on.

Containers can be started, paused, resumed, and stopped, leading many people to draw comparisons with virtual machines (VMs). In reality, though, aside from this similarity, containers are different beasts. Whereas VMs require a host OS, a software abstraction layer known as a hypervisor, and an entire OS installation for each instance, containers are very close to the metal. Each container is just piggybacking on the resources of a single kernel, with just a thin layer of isolation. This means you can run many more containers on a single machine than VMs—they're faster and use less resources.

What Is an Image?

As we just said, a container, in the abstract, is just an empty vessel for executing software in. To start a specific container, you need to supply it with some specific environment or context—what you'd need in order to run an NGINX web server in one container would be quite different from what you'd need to run, say, MySQL in another container.

The environment or context that you supply when you create the container—known as an *image*—is everything that makes the container unique. For example, what does the filesystem look like? What environment variables are set? What command is being run? So, an image is a bundled up package of everything needed to run a (specific) container.

12. https://docs.microsoft.com/en-us/virtualization/windowscontainers/about/